

SYSTEMS BIOLOGY

Quantifying interventional causality by knockoff operation

Xinyan Zhang^{1,2}, and Luonan Chen^{1,3,4,*}

Causal inference between measured variables is crucial to understand the underlying mechanism of complex biological processes at a network level but remains challenging in computational biology. We propose an innovative causal criterion, knockoff conditional mutual information (KOCMI), to accurately infer interventional direct causality without prior knowledge of the network structure using either time-independent or time-series data. KOCMI performs knockoff operation on a variable as its virtual intervention, which preserves the original network structure, and then identifies the causality between two variables by estimating the distributional invariance before and after such a virtual intervention. We show that, algorithmically, KOCMI enables quantification of causal relationship, even for networks with loops, and, theoretically, is also consistent with the do-calculus causal analyses but without their prerequisite of the network structure. KOCMI shows superior performance on benchmark and real datasets, comparing with existing methods. Overall, KOCMI provides a powerful tool in inferring interventional causality, which is theoretically ensured and experimentally validated by real intervention data.

INTRODUCTION

Causal inference from observed data is a fundamental problem in various fields of natural science and engineering, including biology, earth science, economics, medicine, neuroscience, and machine learning. Discovering causal graph of a system is crucial for understanding the system's dynamics and its response to changes in environment. In biological systems, revealing gene regulatory networks (GRNs) or mechanisms is critically important for understanding cellular processes, but constructing a high-quality GRN from observed data remains a substantial challenge in computational biology (1–3). Effectively identifying causal relationships within complex biological systems can offer deeper insights into regulatory mechanisms and clarify biological functions, such as gene regulations, signaling processes, metabolic pathways, and disease progression (4–6). Specifically, deriving a causal regulatory network for a particular disease can uncover the underlying molecular mechanisms, enabling quantitative studies for precise treatment.

With rapid development of high-throughput sequencing technology, many methods have been proposed to infer GRNs. These include model-based approaches such as Boolean networks (7, 8) and Bayesian networks (9, 10), machine learning-based methods like GENIE3 (11) and GRNboost2 (12), information theory-based methods such as CLR (13) and ARACNE (14), and integration-based methods like GENIMS (15) and NIMEFI (16). However, many of them are based on linear or tree-based models, which are hard to be generalized to more comprehensive nonlinear frameworks, and GRNs inferred by these methods are generally undirected networks. Moreover, most of these methods rely on correlation and regression for GRN inference, which remains at the first level of the structural

causal model (SCM) framework, i.e., association, but not real causality (17, 18). The second level of SCM framework is interventional causality (17, 18). Current interventional causal inference methods are primarily based on the do-calculus (18–20), which depends on the prior knowledge of network structure, such as the back-door (20, 21) and front-door criteria (20, 22, 23). However, the prerequisite of known network structure substantially limits their applicability to real biological networks. Also many of such methods are only applicable to inferring an acyclic network, which is not suitable to real-world problems.

Here, to address these problems, we propose an innovative approach to identify interventional direct causality between variable pairs based on either time-independent data or time-series data without prior knowledge of the network structure, i.e., knockoff conditional mutual information (KOCMI), which, theoretically, is also consistent with the do-calculus causal analyses but overcomes their limitation. As a theoretical foundation, we show that the interventional distribution can be obtained just from the observed data in Theorem 1. In particular, KOCMI is also effective to infer a network with loops, in contrast to the acyclic structural requirement of many existing methods. Specifically, KOCMI performs knockoff operation on a variable to generate its knockoff as its virtual intervention but without real experiments while preserving the original network structure (Fig. 1A). It then identifies causal relationships between two variables in an interventional sense and quantifies their causal strength based on distribution invariance before and after such a virtual intervention (Fig. 1B).

KOCMI differs from the original knockoff framework in its inferential focus: Whereas the original method controls the false discovery rate (FDR) at the node level for feature selection, KOCMI targets FDR control at the level of individual directed edges. While KOCMI can be extended to infer entire causal networks by systematically evaluating all variable pairs, its primarily theoretical foundation and statistical guarantees are focused on edge-level inference.

The effectiveness of the KOCMI method has been robustly validated across multiple benchmark datasets and real-world datasets. Furthermore, KOCMI has demonstrated its ability to identify disease genes, and the validation in surgical intervention datasets shows

¹Key Laboratory of Systems Biology, Shanghai Institute of Biochemistry and Cell Biology, Center for Excellence in Molecular Cell Science, Chinese Academy of Sciences, Shanghai 200031, China. ²University of Chinese Academy of Sciences, Beijing 100049, China. ³Key Laboratory of Systems Health Science of Zhejiang Province, School of Life Science, Hangzhou Institute for Advanced Study, University of Chinese Academy of Sciences, Chinese Academy of Sciences, Hangzhou 310024, China. ⁴School of Mathematical Sciences and School of AI, Shanghai Jiao Tong University, Shanghai 200240, China.

*Corresponding author. Email: lncchen@sibcb.ac.cn

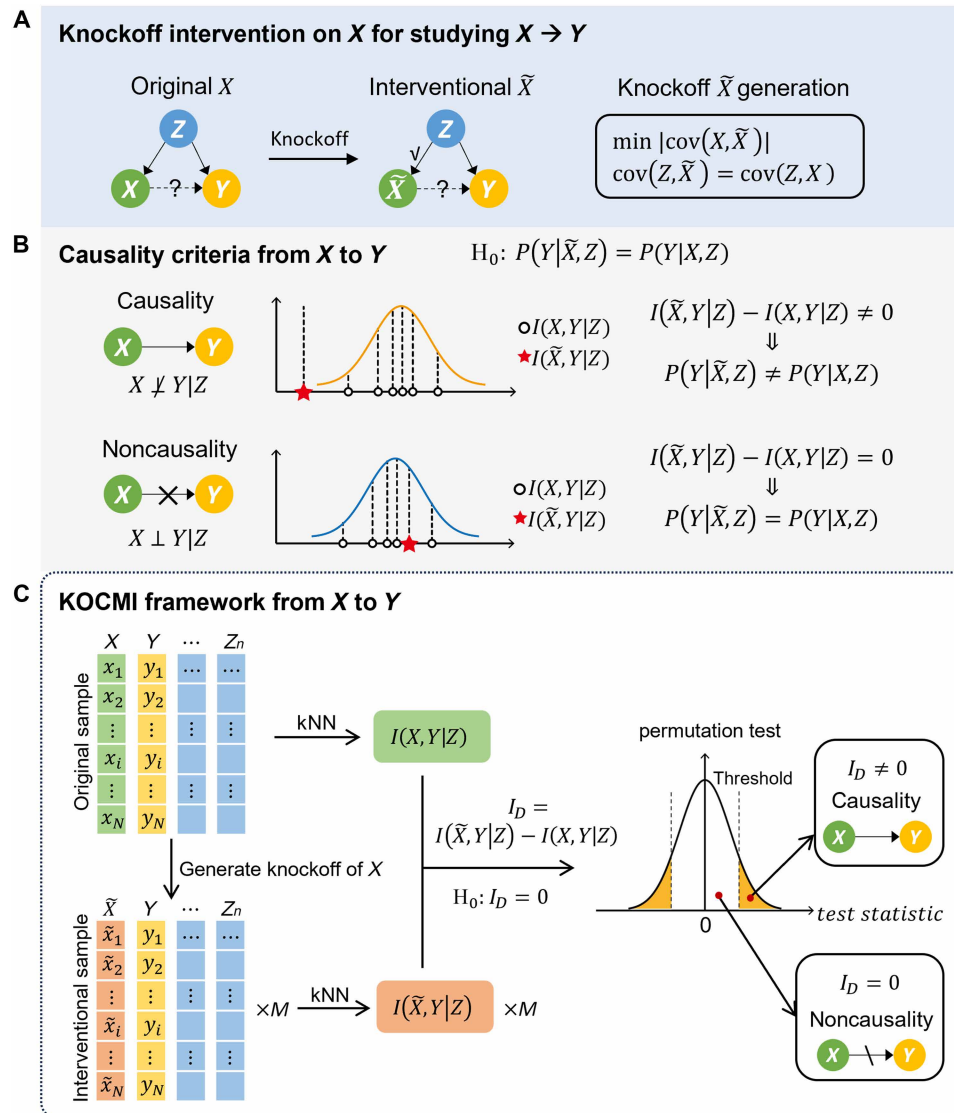


Fig. 1. The framework of KOCMI. (A) Knockoff intervention on X for studying $X \rightarrow Y$: generate a knockoff variable $\tilde{X}$ for the cause variable X . Knockoff method can ensure that correlation between the generated knockoff $\tilde{X}$ and other variables is the same as the correlation between the original variable X and other variables. (B) Distribution invariance is used as the criterion for causal inference: If the conditional distribution of the outcome/effect variable changes after knockoff intervention, then a causal relationship exists; if the conditional distribution remains unchanged, then no causal relationship is present. (C) The computational framework of KOCMI: first, generate M knockoffs of the cause variable. Then, estimate the conditional mutual information (CMI or I) between the outcome variable and the original cause variable, as well as CMI between the outcome variable and the knockoff variable using the kNN method. A hypothesis test is used to determine whether the CMI changes before and after the intervention. If there is a change, then a causal relationship is inferred; if there is no change, then no causal relationship exists. Note: I means CMI, and I_D means $D_{X \rightarrow Y}$.

that the causal relationships identified by KOCMI through virtual interventions are consistent with those observed in real interventions. KOCMI is applicable to various fields as a general method for causal inference and is also freely available in the Supplementary Materials.

RESULTS

Overview of causal inference

Information-based and regression-based methods (24) are the most popular methods in inferring GRN due to low computational complexity and sample (size) demand. However, the causality inferred by such a method, e.g., correction-based or regression-based method,

is still at the first level of the SCM framework, which is association but not real causality (17, 18). The second level of SCM framework is interventional causality (17, 18). Current interventional causal inference methods are mostly based on the do-calculus (18, 19), which depend on the prior knowledge of network structure, such as the back-door (21) and front-door criteria (22, 23).

Here, we proposed an innovative method based on distribution invariance, i.e., KOCMI, with knockoff operation rather than traditional do-calculus, which is able to infer direct interventional causality for both steady-state and time-series data, thus overcoming such limitations. In addition, our method is able to infer a network even with loops, which is also an advantage over many traditional methods.

Performance on benchmark datasets

We benchmarked KOCMI method against seven other well-known network reconstruction methods [GENIE3 (11), nonlinear-ODE (25), GENIMS (15), CLR (13), ARACNE (14), TIGRESS (26), and NIMEFI (16)] using different types of data, including the simulated data, DREAM challenges, biosynthesis network (IRMA data) (27), and further two real data, the SOS DNA repair network in *Escherichia coli* (28, 29) and the human HeLa data (30).

First, we simulated five causal scenarios between three variables, cascade, fan-out, sequential, fan-in, and loop, which can represent all one-way causal scenarios among the three variables (Fig. 2A and fig. S3A). The performance of different methods on the five causal scenarios was illustrated (Fig. 2B and fig. S3B). KOCMI performed better than other methods with high average and small variance of area under the receiver operating characteristic curve (AUROC) and area under the precision-recall curve (AUPR) across the five causal scenarios and maintained robustness with increasing noise levels. With middle noise level, KOCMI had the highest product of AUROC and AUPR on the cascade, fan-out, sequential, and loop scenarios and slightly lower than nonlinear-ODE on the fan-in case (Fig. 2C and fig. S3C). Furthermore, we observed that it was difficult for GENIE3, nonlinear-ODE, and TIGRESS to deal with the case containing loop structures. CLR and ARACNE, which were based on mutual information, performed bad on fan-in case.

The IRMA network (27) is a synthetic network embedded in *Saccharomyces cerevisiae*, widely used as benchmark data in inferential methods. This network contains five genes and eight regulatory edges, whose structure is known, with data obtained from two performed perturbation experiments (switch on and switch off). We applied KOCMI, respectively, on switch-on dataset, switch-off dataset, and their combined dataset, comparing with other methods. The results showed that KOCMI had the highest average of AUROC and AUPR across all the three datasets (Fig. 2D). Specifically, on the switch-on dataset and switch-off dataset, KOCMI performed better than other methods with the highest product of AUROC and AUPR. On the combined dataset, KOCMI achieved the second-highest product of AUROC and AUPR, ranking just below GENIE3 (fig. S3D).

The SOS DNA repair network is a real non-time-series data verified by experiments in *E. coli*, composed of nine genes and 24 regulations (28, 29). By comparing the inference results of the real data, KOCMI had better performance than other methods with 0.59 AUROC and 0.42 AUPR (Fig. 2E). The human HeLa cell cycle dataset is a biological experiment data (30). Sambo *et al.* (31) reported a subnetwork consisting of nine regulatory relationships among nine genes, with six corresponding time-series datasets. KOCMI was applied to each dataset, compared with other methods using AUROC and AUPR metrics. The comprehensive performance of different methods on the six datasets was illustrated (Fig. 2F and fig. S3E). KOCMI outperformed the other methods on each dataset with highest AUROC and AUPR scores. It can be observed that the advantage of KOCMI in AUPR is particularly pronounced, indicating that KOCMI enhances the inference performance for true regulatory relationships, compared to other methods.

DREAM3 and DREAM4 In Silico Network Challenges data, which contain both steady-state and time-series data, were also applied for benchmarking. They are the most popular benchmark datasets used in GRN reconstruction algorithms (32–35). We also investigated the performance of these methods in different data types of DREAM dataset by common evaluation metrics. Each method was applied

separately to steady data, time-series data, and their combination to infer the network. The results from different types of data indicated that KOCMI obtained higher AUROC and AUPR on all five networks at both size_10 DREAM3 and size_10 DREAM4 (Fig. 3A and figs. S4 and S5). Furthermore, KOCMI demonstrated the best overall performance on both static and time-series data (Fig. 3B), indicating its ability to maintain high efficacy across different data types. The average of AUROC and AUPR across all 10 networks from the DREAM3 and DREAM4 challenges further highlight that KOCMI has superior overall performance (Fig. 3C). Taking the network 1 of DREAM4 as an example, the inferred networks of the top three performing methods (KOCMI, TIGRESS, and NIMEFI) are illustrated (Fig. 3D), which had the top three products of AUROC and AUPR. TIGRESS and NIMEFI identified 9 and 12 true-positive regulatory links and 9 and 16 false-positive regulatory links, respectively, while KOCMI identified 12 true-positive links and 11 false-positive links. It is obvious that KOCMI assigned higher weights to edges in the ground-truth network compared with other methods. Additionally, the overall performance of KOCMI on time-series data (DREAM3, DREAM4, and IRMA) outperforms both dynGENIE3 (36) and BINGO (37) (fig. S6). In conclusion, KOCMI could identify the nonlinear relationships and be more sensitive to true regulatory relationship compared to these methods.

Validation of KOCMI using metagenomic data

The skin microbiome is crucial for maintaining skin homeostasis, and disruptions in its balance are linked to the onset and progression of various common skin diseases (38, 39). Here, we applied KOCMI to infer causal relationships between microbial phylum levels and skin physiological indicators. A total of 265 forehead samples and 277 cheek samples were collected (Fig. 4A), with each sample having measurements for skin microbiome and skin physiological indicators. The results of the permutation test indicated that the skin physiological indicators, including transepidermal water loss (TEWL), sebum, pH and porphyrin, had significant differences between the forehead and cheek ($P < 0.05$) (Fig. 4B), which may cause or be caused by the difference of microbiome. KOCMI was applied to this dataset to infer the causal regulatory relationships from the microbiota to skin physiological indicators on the forehead and cheek, respectively. Positive or negative causality is judged by the Pearson correlation coefficient (PCC) (Fig. 4C).

We found that *Actinobacteria* had a positive causal effect to all five indicators in the cheek area, which aligns with the fact that *Actinobacteria* constitutes the largest proportion of the skin microbiome and is associated with various skin diseases (40). *Bacteroidetes* showed a negative causal effect to TEWL, hydration of the cheek, and sebum in both the cheek and forehead areas, consistent with previous studies (40). Furthermore, causal inference at the species level (tables S6 and S7) revealed that *Cutibacterium acnes* had a causal effect to multiple indicators in cheek, particularly sebum, porphyrin, and TEWL, which is in line with existing reports that *C. acnes* induces lipogenesis and production of sebum and is porphyrin-producing strains (41–43). *Staphylococcus* was found to have significant causality to TEWL, pH, and sebum in both areas, consistent with that it can maintain skin homeostasis and prevent opportunistic pathogens from causing disease (44) and that high colonization with *Staphylococcus* contributes to the occurrence of seborrheic dermatitis (45). Besides, *Malassezia* is the most abundant genus in the fungal microflora found on human skin, and it is associated with

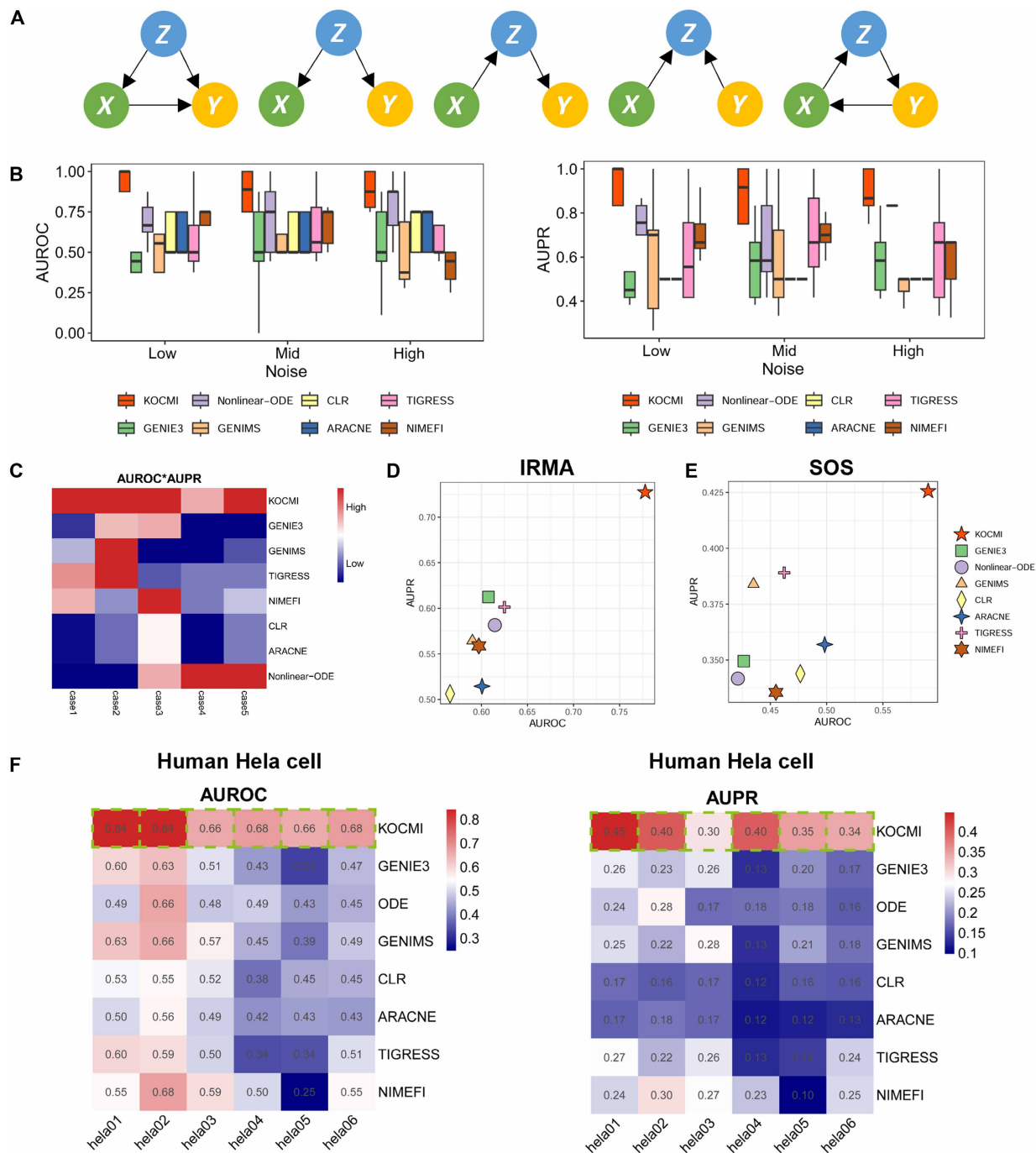


Fig. 2. Performance on benchmark datasets. (A) The five causal scenarios between three variables, cascade, fan-out, sequential, fan-in, and loop. (B) The performances of different methods on the simulated cases with increased noise level. Left is the AUROC values among the five cases, and right is the AUPR values. (C) The product of AUROC and AUPR of different methods on the five cases with middle noise level. (D and E) The performances of different methods on the IRMA dataset and the SOS DNA repair dataset. (F) The performances of different methods on human HeLa cell datasets.

various skin diseases (46, 47). Also, they actually show strong causality from microbes to hydration, porphyrin, and pH.

We further investigated causality from those skin statuses to microbes based on KOCMI because microorganisms are easily influenced by the host environment (fig. S7A). *Actinobacteria* exhibited bidirectional positive causality with multiple indicators, while TEWL and pH were found to have inhibitory effects on Ascomycota. Our analysis

also contributes to select microbes that are common or distinct between the cheek and forehead for further mechanistic investigation.

KOCMI identifies potential key causal genes in cancer

To further demonstrate the biological significance of KOCMI, we reconstructed GRNs using breast cancer data and liver cancer data (48) and conducted a comprehensive analysis, including biomarker

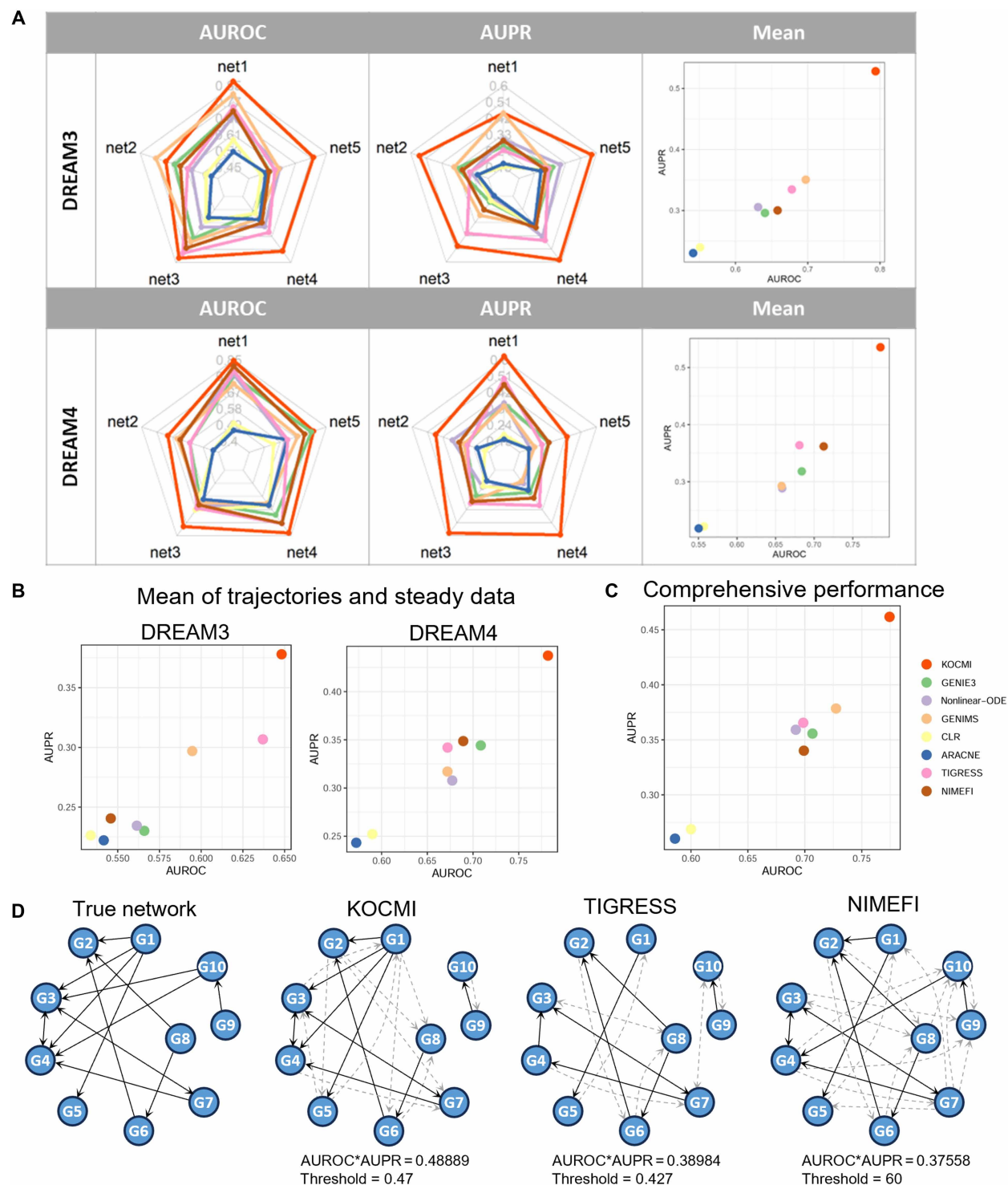


Fig. 3. The performances of different methods on DREAM3 and DREAM4 In Silico Network Challenges data. (A) Performances of different methods on each network in the DREAM3 and DREAM4 challenges, as well as the average AUROC and AUPR across the five networks. (B) The average performance of different methods on trajectories data and steady data. (C) The comprehensive performances of different methods on DREAM3 and DREAM4. (D) The true network of the net1 of DREAM4 and networks reconstructed by KOCMI, TIGRESS and NIMEFI.

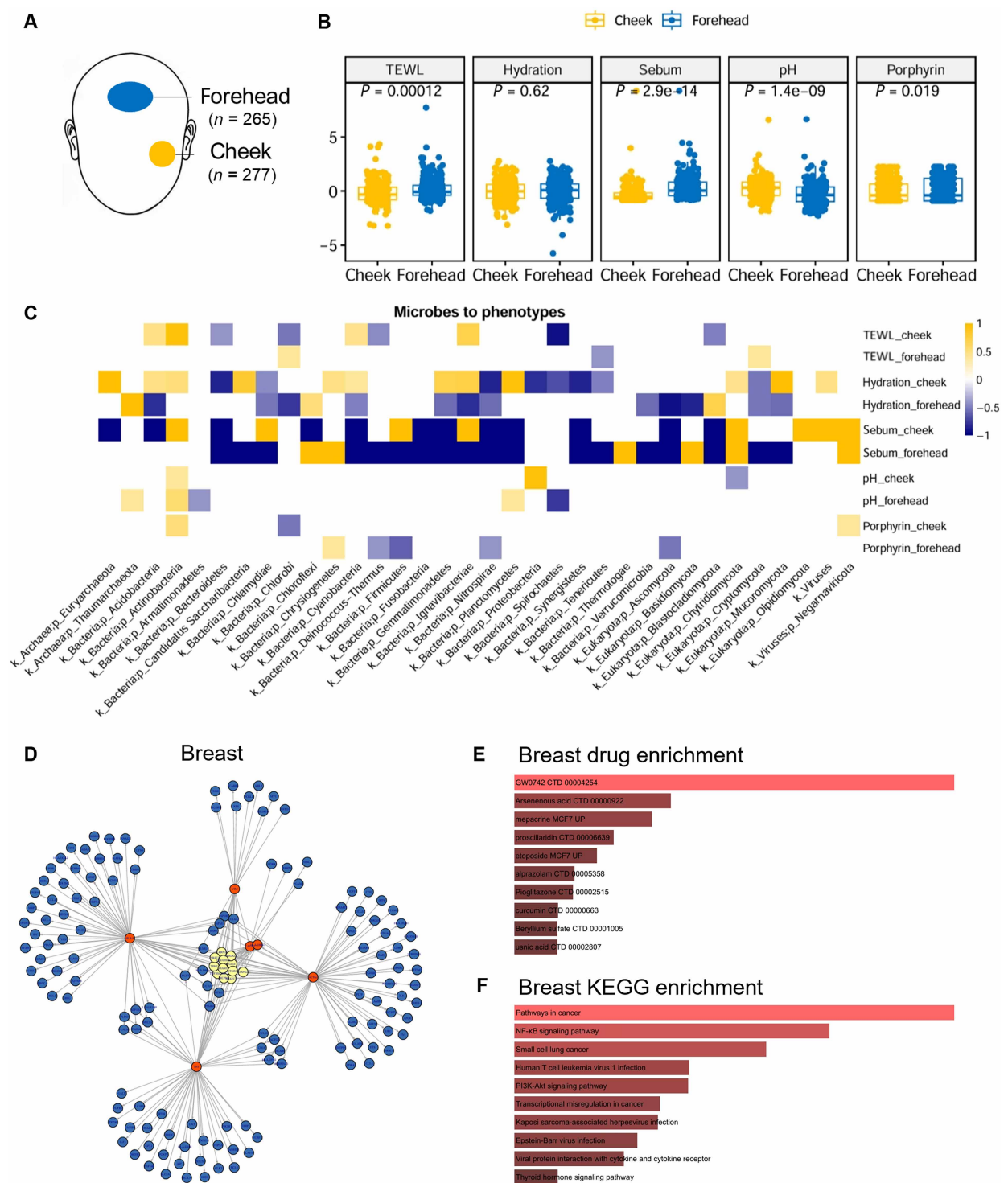


Fig. 4. Validations of KOCMI using metagenomic data and breast cancer data. (A) A total of 265 forehead samples and 277 cheek samples were collected and measured for skin microbiome and skin physiological indicators. (B) The difference of skin physiological indicators between cheek and forehead. (C) The causality identified by KOCMI from microbiota to skin physiological indicators. A positive value represents positive regulation, while a negative value represents negative regulation. The absolute value represents the causal strength. (D) A subnetwork of the network reconstructed by KOCMI using breast cancer data, composing of hub genes and their downstream and shared upstream genes. (E and F) Drug enrichment analysis and Kyoto Encyclopedia of Genes and Genome (KEGG) pathway enrichment analysis for the hub genes and their shared upstream genes. NF-κB, nuclear factor κB; PI3K, phosphatidylinositol 3-kinase.

prediction and enrichment analysis of potential therapeutic drugs. The two cancer datasets have larger sample size (2184 genes and 2220 genes, respectively) and greater research significance. For breast cancer data, KOCMI identified 9600 regulatory relationships among 2184 genes. After reconstructing the GRN, we selected six hub genes with the highest degree, along with their respective downstream genes and shared upstream genes, for visualization (Fig. 4D). Some of the hub genes have already been reported to contribute to breast cancer risk. *HLA-E* is commonly overexpressed in breast cancer cells and contributes to tumor immune escape, specifically natural killer cells, a phenomenon that is likely to have impact on clinical outcome of patients (49, 50). It was reported that a subset of cytotoxic T lymphocytes expressing *HLA-DR* is enriched in breast cancer without axillary lymph node metastasis compared with breast cancer with axillary lymph node metastasis (51). The elevated expression of *VIM* contributes to the aggressive phenotype and poor prognosis in triple-negative breast cancer (TNBC), and *VIM* expression might be useful as a biomarker for the prognosis of TNBC (52). Aberrant *ACTA2* expression accelerated the invasiveness and metastasis of breast cancer cells (53). The shared upstream genes of these hub genes are likely to be potential key genes associated with the occurrence and progression of breast cancer.

Next, Kyoto Encyclopedia of Genes and Genome pathway enrichment analysis was performed on the six hub genes and their downstream genes and their shared upstream genes by Enrichr (<https://maayanlab.cloud/Enrichr/>). Figure 4F shows the top 10 enriched potential pathways, which have varying levels of relevance to cancer, especially breast cancer. The phosphatidylinositol 3-kinase–Akt pathway was one of the most crucial. Dysregulation of this pathway, particularly through mutations in *PIK3CA*, promotes tumor growth, survival, and metastasis (54). The nuclear factor κ B pathway enhanced proliferation and resistance to apoptosis especially in aggressive subtypes like TNBC (55). Furthermore, we performed drug enrichment analysis on the six hub genes and their shared upstream genes. Figure 4E shows the top 10 enriched potential drugs based on Drug Signature Database (DSigDB). Of the 10 drugs listed, four have shown potential or have been researched for breast cancer treatment or drug resistance. 5-Azacytidine was studied for reversing gene silencing in tamoxifen-resistant breast cancer cells (56). Clinical trials have shown the ability of fenretinide to reduce the risk of second breast cancer, particularly in premenopausal women (57). Pyrvinium has been investigated for its ability to inhibit Wnt signaling, a pathway implicated in breast cancer progression (58). Pentoxifylline showed the ability to reduce the proportion of regulatory T cells in the population of breast tumor-infiltrating lymphocytes and modifies the balance of cellular and cytokine responses in favor of antitumor responses (59). This indicates that these genes may serve as promising drug targets for further investigation and treatment of breast cancer that demonstrates the effectiveness of KOCMI. For liver cancer data, we conducted the same analysis, and the results are presented in note S1.

KOCMI identifies key genes in diabetes in a sense of interventional causality

We applied KOCMI to a dataset comprising the time-series data of 19 individuals' recovery process after Roux-en-Y gastric bypass (RYGB) surgery to validate its effectiveness in an interventional sense. In the dataset, the plasma samples and clinical indicators (glucose and cholesterol) were collected from nine normal glucose-tolerant obese (NO) and 10 diabetic obese (DO) patients before (P) and 1 week (1W), 3 months (3M), and 1 year (1Y) after RYGB, and the plasma

samples were obtained at three time points for each individual, including the fasting state and 30 and 45 min after a standardized liquid meal (60). A total of 146 proteins were measured. First, KOCMI was used to identify proteins with causal relationships to clinical indicators based on DO fasting dataset, and Gene Ontology (GO) enrichment analysis was performed on these proteins. For aesthetic clarity in the visualization, the corresponding gene symbols of the proteins were labeled in the figure. In the subsequent text, gene symbols will also be used in place of protein names.

For glucose (Fig. 5A), 12 of the 33 identified proteins were found to be associated with blood glucose regulation in the National Center for Biotechnology Information (NCBI) database, such as IGF2, RBP4, and APOL1 (61, 62). For cholesterol, 14 of the 27 identified proteins were found to be associated with cholesterol regulation in the NCBI database, such as APOC3, APOA1, and LPA (63, 64). The results of the GO enrichment analysis show that proteins with causal relationships to blood glucose are primarily enriched in biological processes such as "lipid transport" and "transforming growth factor- β production" (Fig. 5A). Similarly, proteins with causal relationships to cholesterol are mainly enriched in lipoprotein metabolic processes (fig. S8A), which play a key role in the development and progression of diabetes (65, 66). Differential expression analysis revealed that the expression levels of some identified proteins significantly differ between DO and NO samples, which demonstrates the effectiveness of KOCMI (Fig. 5B and fig. S8B). Due to the small sample size, we relaxed the significance threshold for the *P* value. The positive or negative regulation of proteins to clinical indicators was determined by the Spearman correlation coefficient. In the DO group, proteins identified by KOCMI as positive regulators to glucose, such as FN1, HP, CNDP1, VWF, and SAA2-SAA4, exhibited a gradual decrease in expression following surgical intervention (Fig. 5C and fig. S10). Conversely, negative regulators like IGF2, HRG, APOA4, and LPA showed a gradual increase in expression after surgery, consistent with the decreasing in blood glucose levels (Fig. 5C and fig. S10). This indicated that the causality identified by KOCMI through virtual interventions was consistent with the observation from actual experiment intervention, demonstrating the effectiveness of KOCMI in identifying interventional causality.

We further reconstructed the two protein-protein interaction networks for the preoperative phase of the DO group and the NO group, respectively. The top 20 proteins with the largest degree differences between the two networks were selected for GO functional enrichment analysis and drug enrichment analysis (fig. S8, C to F). These proteins are primarily enriched in functions and components related to the development of diabetes. Additionally, some of the drugs identified in the enrichment analysis have already been proven effective in treating diabetes and its complications (67–69).

Furthermore, we reconstructed protein-protein interaction networks for each individual and calculated the degree of every protein within these networks. Then, 10 proteins with significant degree differences between the DO and NO groups were identified at the network level ($P < 0.1$) (fig. S9A). The expression levels of these 10 proteins were not significantly different between the DO and NO samples (fig. S9B), suggesting that these proteins may influence the development of diabetes primarily through network regulatory interactions. This also confirms that network-level analysis captures information that may not be found at the expression level. We visualized the subnetworks composed of these 10 proteins and their upstream and downstream interactions (Fig. 5D). Among these

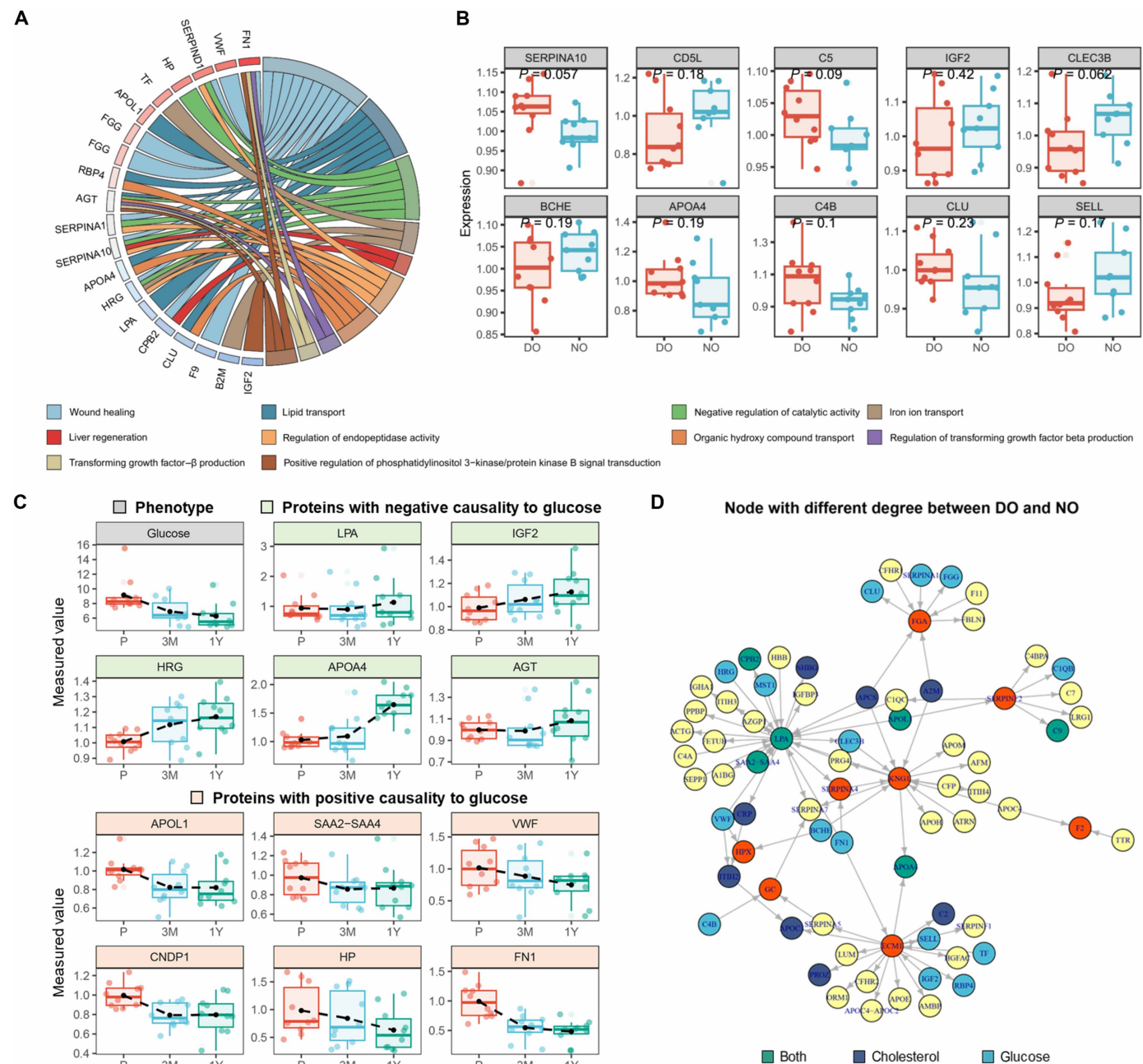


Fig. 5. KOCMI identifies interventional causality in diabetes. (A) The causal proteins identified by KOCMI to glucose and Gene Ontology (GO) enrichment analysis. (B) The differences in expression of the proteins identified by KOCMI between the DO group and the NO group. (C) The changes in protein expression following real surgical intervention are consistent with the regulatory relationships to glucose identified by KOCMI. (D) Regulatory networks consisting of the proteins with significant degree differences between the individual protein-protein interaction (PPI) networks of the DO and NO groups, along with their upstream and downstream proteins, including many proteins identified by KOCMI as having causal effects on glucose or cholesterol.

10 proteins, two (LPA and ITIH2) were previously identified as having causal relationships with blood glucose and cholesterol. Many of their upstream and downstream proteins were also previously identified as having causality to blood glucose and cholesterol. This further suggests that these proteins influence the development of diabetes through regulatory interactions, reinforcing the validity of the KOCMI.

DISCUSSION

In our study, we present an innovative approach KOCMI, which theoretically is consistent with the do-calculus causal analysis to accurately identify interventional direct causality but without prior knowledge of the network structure. In particular, we theoretically show that the interventional distribution can be derived from the observed or uninterventional data in Eq. 3. On the basis of the

knockoff theory, we make knockoff operation on a variable to generate its knockoff variable, which has the same correlations with other variables except the potential effect variable, as a virtual intervention. The virtual intervention achieved by replacing the original variable with knockoff variable does not cut off the relations between the original variable and others. Therefore, the knockoff operation can intervene on a variable while maintaining the original network structure. This overcomes the limitation of many causal inference methods that require prior knowledge of the network structure. In addition, KOCMI can be applied to infer a network with feedback loops, thus having wider applications to various fields. Next, the distribution invariance before and after such virtual intervention is used as the criterion for identifying the causality. Conditional mutual information is used to numerically estimate the differences between the distributions. Hypothesis testing is used to assess the significance of the differences. KOCMI was compared with seven other popular causal inference methods across multiple benchmark datasets, demonstrating its superior performance. Applications on various biological datasets with real interventions as well as identification of disease causal genes validated the effectiveness of our method in the sense of interventional causality and also highlighted the biological significance of KOCMI. It is important to clarify that KOCMI differs from the original knockoff filter in its inferential target. While the original knockoff framework controls FDR at the node level (i.e., identifying variables associated with a specific outcome node), KOCMI controls FDR at the edge level (i.e., for individual directed edges). Although KOCMI can be extended to reconstruct causal networks by systematically testing all variable pairs, its theoretical foundation and statistical guarantees are focused on edge-level inference. Moreover, KOCMI remains effective even in low-dimensional scenarios (e.g., with only two or three nodes), where the original knockoff filter may lose statistical power. This makes KOCMI broadly applicable across both high- and low-dimensional settings.

Of course, there are still some shortcomings here. Our method is valid under the assumption of no unobserved confounders. Besides, we observed that, as the network scale increases, the method faces challenges, such as heightened computational complexity because inferring the network requires iterating through each node as effect variable and applying our method. Additionally, the methods used for numerically estimating mutual information can also affect the accuracy of KOCMI. Now, KOCMI is primarily designed to control the FDR at the level of individual directed edges. When reconstructing causal networks, we recommend controlling the global FDR using adjusted P values and q values. In addition, we generate knockoffs for single variable to identify one-to-one causal relations, but knockoff theory allows for the generation of knockoffs for multiple variables simultaneously, which would enable us to identify many-to-one causal relationships. Besides, knockoff operation is model free, so it can be applied in various frameworks, containing deep learning. In the future, we will explore ways to improve our method to ensure its robustness even in the presence of unmeasured confounders. We plan to explore more effective and principled approaches for graph-level FDR control. In addition, we hope to improve the accuracy of the method by generating knockoffs using deep learning framework and also expand the method to identify many-to-one causal relationships for complex high-order causal analysis of biological systems.

MATERIALS AND METHODS

Knockoff operation

The knockoff framework was developed by Barber and Candès in 2005 (70, 71) as a tool for controlling FDR. Given a set of observed n variables $\mathbf{X} = (X_1, X_2, \dots, X_n)$, the knockoff variables of the observed variables, defined as $\tilde{\mathbf{X}} = (\tilde{X}_1, \tilde{X}_2, \dots, \tilde{X}_n)$, are constructed to be in-distribution null variables and as decorrelated as possible from the original data. The original and knockoff variables are exchangeable: Taking any subset of variables and swapping them with their knockoffs leave the joint distribution invariant. This is accomplished by ensuring that correlation between the generated knockoff variable and others is the same as the correlation between the original observed variable and others. Then, we have the knockoff operation as follows.

Definition 1. Given a set of original n variables $\mathbf{X} = (X_1, X_2, \dots, X_n)$, their n knockoff variables $\tilde{\mathbf{X}} = (\tilde{X}_1, \tilde{X}_2, \dots, \tilde{X}_n)$ are constructed with the following two properties:

- 1) for any subset $A \subset \{1, \dots, n\}$, $(\mathbf{X}, \tilde{\mathbf{X}})_{\text{swap}(A)} \stackrel{d}{=} (\mathbf{X}, \tilde{\mathbf{X}})$;
- 2) $\tilde{\mathbf{X}} \perp Y | \mathbf{X}$ if there is a response Y ,
where $\stackrel{d}{=}$ represents equal distributions and $(\mathbf{X}, \tilde{\mathbf{X}})_{\text{swap}(A)}$ is obtained from $(\mathbf{X}, \tilde{\mathbf{X}})$ by swapping the entries X_j and $\tilde{X}_j$ for each $j \in A$.

The property (2) is guaranteed if $\tilde{\mathbf{X}}$ is constructed with looking at Y .

There is an example (71) of a joint distribution obeying the two properties for Gaussian variables $\mathbf{X} \sim N(0_n, \Sigma)$

$$(\mathbf{X}, \tilde{\mathbf{X}}) \sim N(0_{2n}, G), G = \begin{bmatrix} \Sigma & \Sigma - \text{diag}\{s\} \\ \Sigma - \text{diag}\{s\} & \Sigma \end{bmatrix} \quad (1)$$

where $\text{diag}\{s\}$ is any diagonal matrix selected in such a way that the joint covariance matrix G is positive semidefinite, and Σ is the covariance matrix of the original $\mathbf{X}$. Here, $\text{diag}\{s\}$ is to be maximized and represents the difference between the original $\mathbf{X}$ and its knockoff version $\tilde{\mathbf{X}}$. As a result, the off-diagonal term $\Sigma - \text{diag}\{s\}$ is minimized, making the covariance of the original $\mathbf{X}$ with its corresponding knockoff version as low as possible. Then, the knockoffs $\tilde{\mathbf{X}}$ can be drawn from the conditional distribution

$$P_{\tilde{\mathbf{X}}|\mathbf{X}}(\tilde{\mathbf{X}}|\mathbf{X}) = N[(\mathbf{X} - \mathbf{X}\Sigma^{-1}\text{diag}\{s\}), (2\text{diag}\{s\} - \text{diag}\{s\}\Sigma^{-1}\text{diag}\{s\})] \quad (2)$$

Conceptually, a knockoff variable is constructed by maximizing the difference between the knockoff variable and its original variable while preserving its relationship with all other variables, as a virtual intervention or knockoff operation on this variable.

Here, we assumed that the joint distribution of the random variables $\mathbf{X} = (X_1, X_2, \dots, X_n)$ follows a joint Gaussian distribution, and there are no unmeasured confounders, which enables us to generate knockoff variables by Eq. 2.

In causal inference, knockoff operation is consistent with the do-calculus causal analysis. By back-door or front-door criteria, the distribution with do-calculus can be estimated from observed data. Similarly, the distribution with knockoff operation can also be estimated from observed data, but without necessity to satisfy back-door or front-door criteria. Next, we consider the causal relation from X to Y

in an unknown network with n observed variables $\mathbf{X} = (X_1, X_2, \dots, X_n)$, including variable X ($X = X_i$), variable Y ($Y = X_j$), and other remaining $(n - 2)$ variables as a set of Z ($Z = \mathbf{X}/\{X_i, X_j\}$), and, then, we have the following theoretical result.

Theorem 1. When considering the causal relation from X to Y by taking all variables other than (X, Y) as a set of variables Z , knockoff variable $\tilde{X}$ (of the original variable X) is constructed without looking at Y , and, then, the distribution of Y conditional on X after the knockoff operation is given by the formula

$$P(Y = y | \tilde{X} = x') = \sum_z \sum_x P(Y = y | X = x, Z = z) P(\tilde{X} = x' | X = x', Z = z) P(Z = z | X = x') \quad (3)$$

which was proved in note S1.

This theorem or Eq. 3 implies that we can obtain the interventional distribution just from the observed (uninterventional) data. Based on Eq. 3, KOCMI compares the distribution of Y conditional on interventional X , i.e., $P(Y | \tilde{X})$ with the distribution of Y conditional on original X , i.e., $P(Y | X)$. This is consistent with the concept of comparing $P[Y | \text{do}(X = 0)]$ with $P[Y | \text{do}(X = 1)]$, but without prerequisite of network structure as in the do-calculus analysis, and thus makes KOCMI to have wider applications in various fields. Moreover, KOCMI is effective in a network even with cycles, in contrast to acyclic network requirement of many existing causal approaches.

KOCMI framework

We aim to infer the causal relation from variable X to variable Y given the remaining variables as a set of Z , i.e., assuming that X is a cause variable and Y is an effect/outcome variable. Here, we propose an innovative method termed KOCMI, which uses knockoff operation and distribution invariance to infer interventional direct causality. Based on the original data, KOCMI first makes a knockoff operation on a cause variable X , which is taken as its corresponding knockoff variable $\tilde{X}$. This knockoff variable has the same distribution as the original X and maintains the correlation to other variables except Y . Therefore, the virtual intervention achieved by replacing the original cause variable with its knockoff variable does not change the relations/associations between the cause variable and other variables except Y (Fig. 1A). Then, KOCMI estimates the difference between the original distribution of the effect variable Y conditional on the original cause variable X and the alternative distribution of Y conditional on the knockoff variable $\tilde{X}$ (Fig. 1B). A significant difference indicates that the cause variable affects the distribution of the effect one, suggesting the presence of a causal relationship between the two variables. Conversely, a nonsignificant difference implies the absence of a causal relationship between them.

Specifically, suppose that there are three random variables X , Y , and Z with N original data/samples, i.e.,

$$X = x_i, Y = y_i, \text{ and } Z = z_i, i = 1, \dots, N \quad (4)$$

which will be used to infer the direct causal relationship from X to Y . Based on the data of Eq. 4, the original conditional mutual information (CMI or I) can be computed by Eq. 5

$$\text{CMI}(X, Y | Z) = \sum_z \sum_x \sum_y P(x, y, z) \log \frac{P(y | x, z)}{P(y | z)} \quad (5)$$

where $P(\cdot)$ is the probability distribution of the corresponding random variable/vector. By the knockoff method, we can generate a knockoff variable $\tilde{X}$ of X , i.e., $\tilde{X} = \tilde{x}_i, (i = 1, \dots, N)$, where $\tilde{x}_i$ is the knockoff data of the i th sample of X . Z blocks not only the back-door paths of X but also the indirect paths from X to Y , and, then, conditioning on Z makes the estimated causality a direct causality. Thus, we can quantify the interventional direct causality from X to Y by designing a measure as Eq. 6

$$D_{X \rightarrow Y} = \text{CMI}(\tilde{X}, Y | Z) - \text{CMI}(X, Y | Z) = \sum_z \sum_x \sum_y \sum_{\tilde{x}} P(x, \tilde{x}, y, z) \log \frac{P(y | \tilde{x}, z)}{P(y | x, z)} \quad (6)$$

Then, we consider two possible values of $D_{X \rightarrow Y}$: (i) $D_{X \rightarrow Y} = 0 \Rightarrow P(y | \tilde{x}, z) = P(y | x, z)$ and (ii) $D_{X \rightarrow Y} \neq 0 \Rightarrow P(y | \tilde{x}, z) \neq P(y | x, z)$.

Theoretically, when X has no direct causal effect on Y , i.e., $X \perp Y | Z$, then

$$P(Y | X, Z) = P(Y | X, \tilde{X}, Z) = P(Y | \tilde{X}, Z)$$

where the first equality holds because $\tilde{X} \perp Y | (X, Z)$, a property guaranteed by the knockoff construction. The second equality holds due to the assumption $X \perp Y | Z$.

Conversely, when X has a direct causal effect on Y , i.e., $X \not\perp Y | Z$, then

$$P(Y | X, Z) = P(Y | X, \tilde{X}, Z) \neq P(Y | \tilde{X}, Z)$$

Therefore, the difference between $P(Y | X, Z)$ and $P(Y | \tilde{X}, Z)$ reflects a direct causal effect from X to Y .

Definition 2. (Distribution invariance) X and Y are two random variables, and Z is the third variable or variable set. If the intervention on X affects the probability distribution of Y under the condition Z , i.e., $P(Y | \tilde{X}, Z) \neq P(Y | X, Z)$, then there is direct causality from X to Y . Conversely, if the intervention on X does not affect the probability distribution of Y under the condition Z , i.e., $P(Y | \tilde{X}, Z) = P(Y | X, Z)$, then there is no direct causality from X to Y .

To infer the direct causality from X to Y , a hypothesis testing can be established based on distribution invariance: $H_0: P(Y | \tilde{X}, Z) = P(Y | X, Z)$; $H_1: P(Y | \tilde{X}, Z) \neq P(Y | X, Z)$. When H_0 is accepted, no direct causality from X to Y exists. Conversely, when H_1 is accepted, a direct causality from X to Y is present.

Therefore, with Eq. 6, an alternative criterion for interventional direct causality from X to Y given a variable set Z (all other variables) was designed as follows:

(i) If $D_{X \rightarrow Y} = 0$, then $P(Y | \tilde{X}, Z) = P(Y | X, Z)$, and there is no direct causality from X to Y ;

(ii) If $D_{X \rightarrow Y} \neq 0$, then $P(Y | \tilde{X}, Z) \neq P(Y | X, Z)$, and there is direct causality from X to Y .

Practically, random errors affect the accuracy of determining whether $D_{X \rightarrow Y}$ is zero or not based on a single sample. According to GhostKnockoff method (72), M knockoffs $\tilde{X}^{(m)}$ ($m = 1, \dots, M$) for X can be generated simultaneously. Every knockoff will be used to

compute a $D_{X \rightarrow Y}$, then $D_{X \rightarrow Y}$ will be computed M times, which can be viewed as M samples from one distribution. Therefore, whether $D_{X \rightarrow Y}$ is significantly different from zero or not can be determined by a hypothesis test ($H_0: D_{X \rightarrow Y} = 0; H_1: D_{X \rightarrow Y} \neq 0$), which can reduce the false-positive rate of causal inference caused by random errors.

Definition 3. X and Y are two variables, and Z is the third variable or variable set. The causality strength from X to Y is defined as $cs = |\overline{D}_{X \rightarrow Y}| / \text{sd}(D_{X \rightarrow Y})$.

Numerical quantification of KOCMI

The M knockoffs $\tilde{X}^{(m)}$ ($m = 1, \dots, M$) of X can be numerically generated from the GhostKnockoff method (72) (R package “GhostKnockoff,” R version 4.3.2).

Many algorithms (73–78) have been developed to estimate the entropy from data. Here, we use k-nearest neighbors (kNN) estimation method (73), which is a suitable nonparametric approach for entropy estimation of high dimensional data, to estimate the conditional mutual information $\text{CMI}(X, Y | Z)$ or $\text{CMI}(\tilde{X}, Y | Z)$ based on Eq. 7 (79), e.g.,

$$\widehat{\text{CMI}}(X, Y | Z) = \psi(k) - \langle \psi(n_{xz} + 1) + \psi(n_{yz} + 1) - \psi(n_z + 1) \rangle \quad (7)$$

where n_A is the number of neighbors of the reference point in the space A with distance smaller than the distance to the k th neighbor of the given reference point in the joint space (X, Y, Z) . It is generally not necessary that the metric is the same in all of the spaces, but we have elected to use the maximum norm in all cases.

Thus, we can obtain the M estimations of $D_{X \rightarrow Y}$ through computing $\widehat{\text{CMI}}(X, Y | Z)$ and $\widehat{\text{CMI}}(\tilde{X}^{(m)}, Y | Z)$ ($m = 1, \dots, M$), which are used to establish hypothesis test ($H_0: D_{X \rightarrow Y} = 0; H_1: D_{X \rightarrow Y} \neq 0$). As the M knockoff variables are not independent, being generated from the same data, we use a permutation test, a nonparametric approach that does not assume independence. In addition, $cs = |\overline{D}_{X \rightarrow Y}| / \text{sd}(D_{X \rightarrow Y})$ is defined as the causality strength from X to Y . Furthermore, in small-sample settings (sample size of < 750), we recommend incorporating the additional criterion $cs > \text{threshold}$ as a stricter condition for declaring a causal relationship to further control FDR. We provide recommended threshold values at a 0.05 significance level for different sample sizes (fig. S2D). These values are derived from the 95th percentile of the empirical distribution of the cs statistic under the null hypothesis, obtained via numerical simulations. The details are shown in note S1.

Above all, to infer the interventional direct causality from X to Y , the KOCMI algorithm can be described as follows (Fig. 1C, where “ I ” means CMI, and “ I_D ” means $D_{X \rightarrow Y}$).

Step 1. Generate M knockoffs $\tilde{X}^{(m)}$ ($m = 1, \dots, M$) of X based on X and all other variables Z except Y .

Step 2. Estimate the difference between original CMI and the knockoff CMI: $\hat{D}_{X \rightarrow Y} = \widehat{\text{CMI}}(\tilde{X}, Y | Z) - \widehat{\text{CMI}}(X, Y | Z)$ for each generated knockoff.

Step 3. Determine whether or not $\hat{D}_{X \rightarrow Y}$ is significantly different from zero by a permutation test ($H_0: D_{X \rightarrow Y} = 0; H_1: D_{X \rightarrow Y} \neq 0$) for M estimations and compute the $cs = |\overline{D}_{X \rightarrow Y}| / \text{sd}(D_{X \rightarrow Y})$ as the causality strength. In small-sample settings, causality can be inferred by combining this measure with the criterion $cs > \text{threshold}$.

In practical application, for causal network inference, one needs to cycle through all variables as Y and to run this method to evaluate their causal relationships with others. Note that KOCMI controls the FDR at the level of individual directed edges; however, when applied to network-level causal inference, additional adjustment for multiple comparisons is required. To address this, KOCMI controls the overall network-level FDR by computing adjusted P values and q values. The pseudocode of KOCMI is provided in note S1.

Supplementary Materials

The PDF file includes:

Supplementary Notes S1 to S3

Legend for code S1

Figs. S1 to S10

Tables S1 to S8

References

Other Supplementary Material for this manuscript includes the following:

Code S1

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